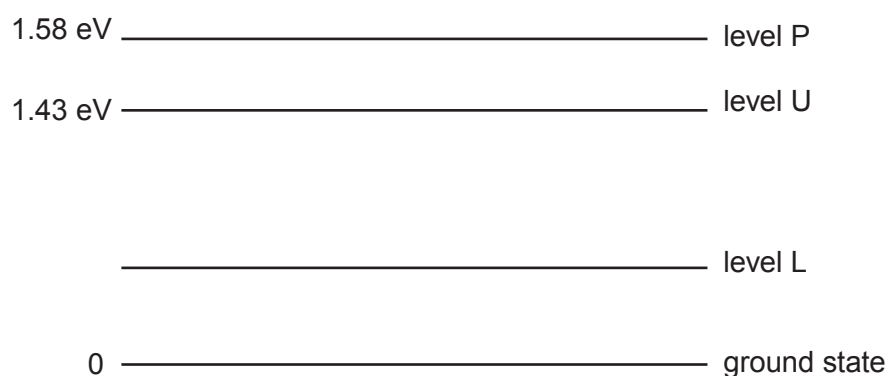


8. The following diagram shows the energy levels for the amplifying medium in a 4-level gas laser.



- (a) The laser transition between level U and level L produces stimulated photons of wavelength $1.06 \times 10^{-6} \text{ m}$. Determine the value of energy level L in **eV**. [3]

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- (b) When explaining the operation of this laser, a student writes the following:

Pumping is required to move electrons from ground level to level P which is a metastable state. This creates a population inversion between P and ground which is needed for the laser to work.

State **two** mistakes in the student's explanation. [2]

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- (c) Semiconductor lasers are regularly used in DVD players and bar code scanners. State **two** advantages that these semiconductor lasers have over gas lasers. [2]

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- (d) As electronic technology advances, outdated electronic devices and their components are discarded. It has been estimated that 50 million tonnes of electronic waste is produced each year and roughly 80% of this is finding its way into landfill. Discuss the effects this electronic waste may have on society. [2]

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END OF PAPER

